PAGE 3 · HERO LOGICAL ARCHITECTURE

How PeX Resolves a Payment Exception

From exception event to governed resolution and final business outcome.

ZONE 1 · RESOLUTION INTELLIGENCE

1. Ingest & Normalize
Maps raw MT/MX to canonical PaymentExceptionCase

2. 41-Profile Classify
Derives root-cause exception profile & missing fields

3. Evidence Corroboration
Queries DDA, Sanctions, KYC, & SWIFTRef ICDs

4. AI Model Gateway
Generates advisory ReasoningResult JSON & scores

ZONE 2 · DETERMINISTIC CONTROL

5. Deterministic Rules
100% scheme validation — Rules ALWAYS override AI

6. Sanctions Re-Screening
Automated watchlist re-check on proposed repairs

7. Operator Review (HITL)
Side-by-side evidence, rules, & AI rationale review

8. Maker-Checker Approval
Hard $0 SoD limit; Dual-Auth for transfers ≥$250k

ZONE 3 · CONTROLLED EXECUTION

9. Egress Contract Gen
Generates schema-compliant ISO pacs.008 amendment

10. Clearing Dispatch
Dispatches via bank rail; logs transport ACK (DISPATCHED)

11. Final Settlement
Settles ONLY upon pacs.002 ACCP or DDA confirmation

12. Merkle Audit Sealing
Cryptographically seals complete lifecycle into WORM log

CORE ARCHITECTURAL MANTRA: **AI recommends. Controls validate. Humans authorize. Systems execute.**

Closed-Loop Governance: EVENT → INTELLIGENCE → DETERMINISTIC CONTROL → HUMAN AUTHORIZATION → EXECUTION → SETTLEMENT → MERKLE AUDIT